

MSP430G2231-based Temperature Indicator and Controller

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The ambient temperature must be within certain limits for instruments to work properly. Therefore temperature is the most measured process variable in industrial automation. Temperature indicators and controllers are now becoming common even in home appliances. For instance, air-conditioners have built-in temperature indicators.

The ambient temperature at any place keeps varying during different times of the day and night. So here we describe a project based on MSP430 microcontroller that indicates the ambient temperature and controls home appliances such as a cooler or a fan at a predetermined temperature. Three 7-segment displays are used to display the temperature.

Main features of this controller are:

1. It also indicates the temperature.
2. It employs MSP430 with internal temperature sensor.
3. The MSP430 is a small but 16-bit device.
4. Low power consumption
5. Development tools are easily available.

This temperature indicator and controller uses minimal components. It is so simple that even a hobbyist can build it without much effort. No special tools are needed to build this project. All the ICs used here are available in DIP packages. So no special placement tools are needed. The cost of components is minimal and parts are commonly accessible.

Circuit description

Fig. 1 shows the block diagram of the temperature indicator and controller using MSP430G2231 microcontroller, 7-segment displays, CD4511 display

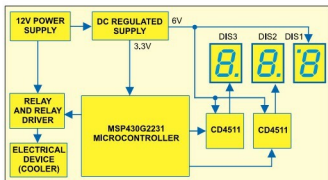


Fig. 1: Block diagram of low-power temperature indicator and controller using MSP430G2231 microcontroller

drivers, power supply, relay and relay driver. The circuit is shown in Fig. 2.

MSP430G2231 microcontroller. The heart of the circuit is a 14-pin MSP430G2231 microcontroller. Texas Instruments' MSP430 family of ultra-low-power microcontrollers consists of several devices featuring different sets of peripherals targeted for various applications. The architecture, combined with five low-power modes, is optimised to achieve extended bat-

tery life in portable measurement applications. The device features a powerful 16-bit RISC CPU, 16-bit registers and constant generators that contribute to maximum code efficiency.

The digitally controlled oscillator (DCO) allows wake-up from low-power modes to active mode in less than 1 μ s.

The MSP430G2x1/G2x31 series is an ultra-low-power mixed-signal microcontroller with a built-in 16-bit timer and ten input/output (I/O) pins. The MSP430G2x31 family members have a 10-bit analogue-to-digital converter (ADC) and built-in communication capability using synchronous protocols (SPI or I²C). For configuration details you may refer to the datasheet. Typical applications include low-cost sensor systems that capture analogue signals, convert them into digital values and then process the data for display or transmission to a host system.

Power supply. The circuit requires a 12V DC supply. For this, a 230V AC primary to 12V, 250mA secondary transformer is used. The bridge rectifier rectifies the AC signal and the capacitor further filters it before feeding to 7806 regulator IC. The 12V supply is directly given to the relay for switching on/off the device. The 6V output from 7806 voltage regulator is used to drive the display section. This 6V is also reduced to 3.3V using zener diode ZD1 to drive the MSP430G2231. LED1 indicates the presence of power supply in the circuit.

PARTS LIST

Semiconductors:	
IC1	- MSP430G2231 microcontroller
IC2, IC3	- CD4511 BCD-to-7-segment display driver
IC4	- 7806, 6V regulator
T1	- SL100 npn transistor
ZD1	- 3.3V zener diode
LED1, LED2	- 5mm light-emitting diode
D1	- 1N4007 rectifier diode
DIS1	- LT542 common-anode, 7-segment display
DIS2, DIS3	- LT543 common-cathode, 7-segment display
BR1	- 1A bridge rectifier module
Resistors (all 1/8-watt, ± 5 per cent carbon):	
R1	- 680-ohm
R2	- 47-ohm
R3	- 1-k Ω -ohm
R4-R18	- 680-ohm
R19, R20	- 470-ohm
Capacitors:	
C1	- 1000 μ F, 35V electrolytic
C2	- 0.1 μ F ceramic disk
C3	- 10 μ F, 16V electrolytic
Miscellaneous:	
X1	- 230V AC primary to 12V, 250mA secondary transformer
S1	- Push-to-on switch
RL1	- 12V, 1C/O relay

